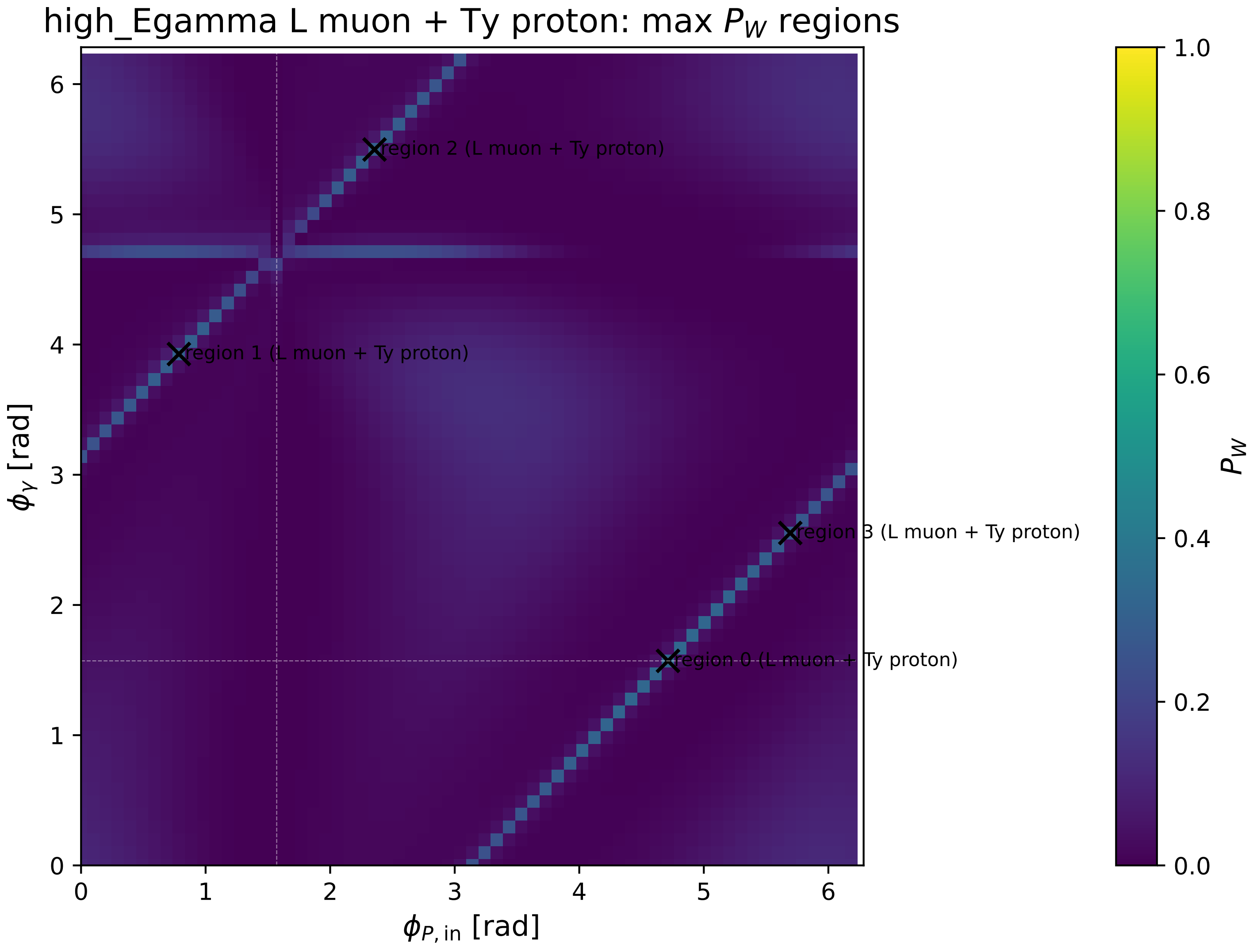
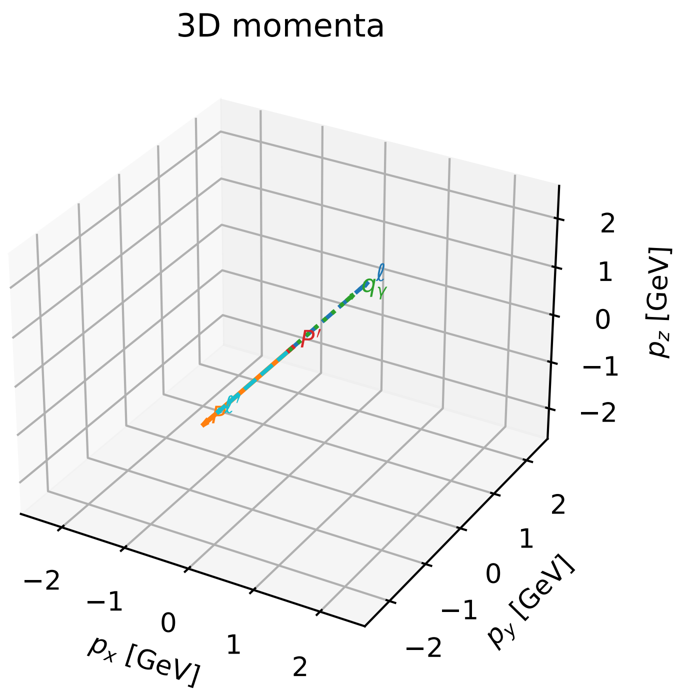

high_Egamma L muon + Ty proton: max P_W regions
region 2 (L muon + Ty proton)
region 1 (L muon + Ty proton)
region 3 (L muon + Ty proton)
region 0 (L muon + Ty proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
P_W
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

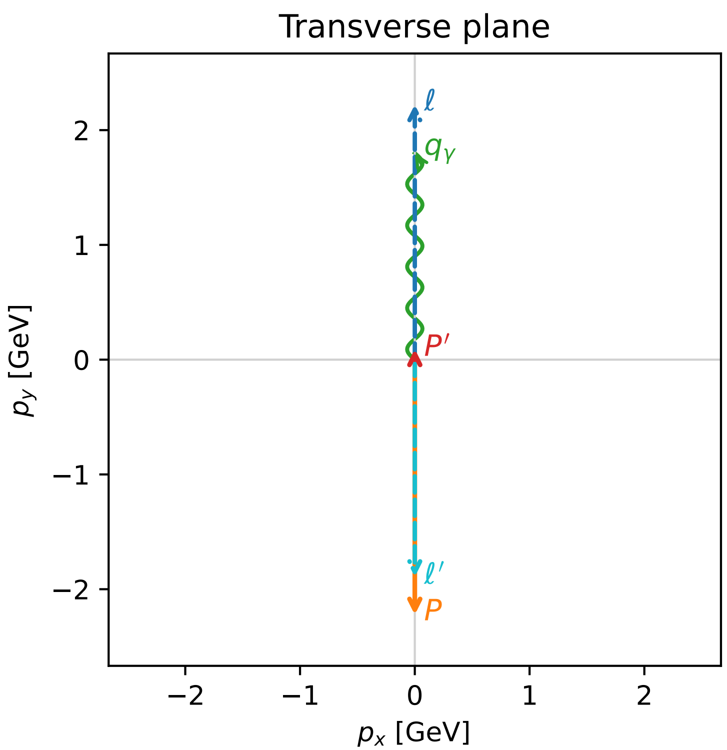
L muon + Ty proton characteristic max P_W configuration

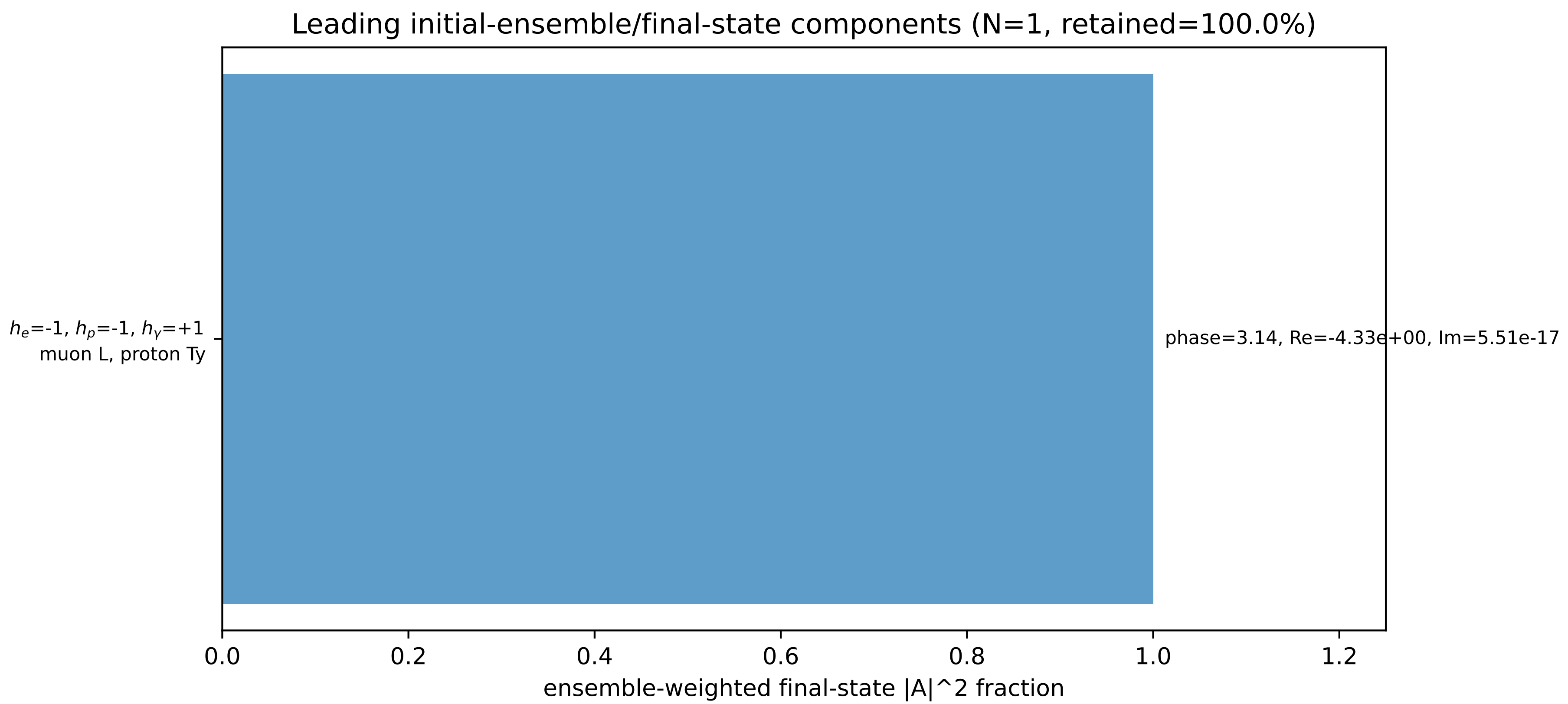


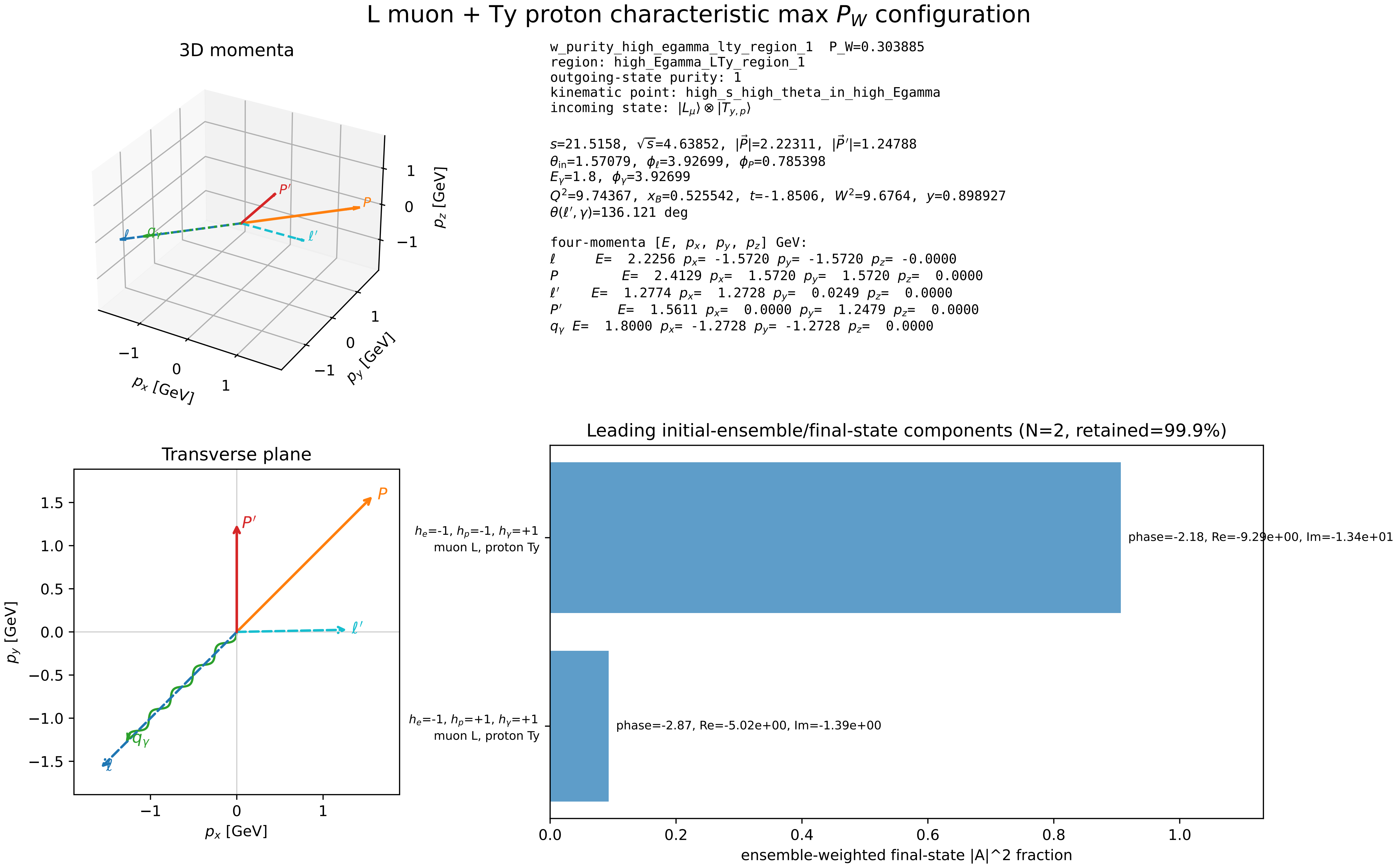
w_purity_high_egamma_lty_region_0 P_W=0.333333
region: high_Egamma_LTy_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{y,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=0.0929745
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1.8, ϕ_γ=1.5708
Q^2=16.8335, x_B=0.846238, t=-3.20247, W^2=3.93848, y=0.964473
$\theta(\ell', \gamma)$=180 deg

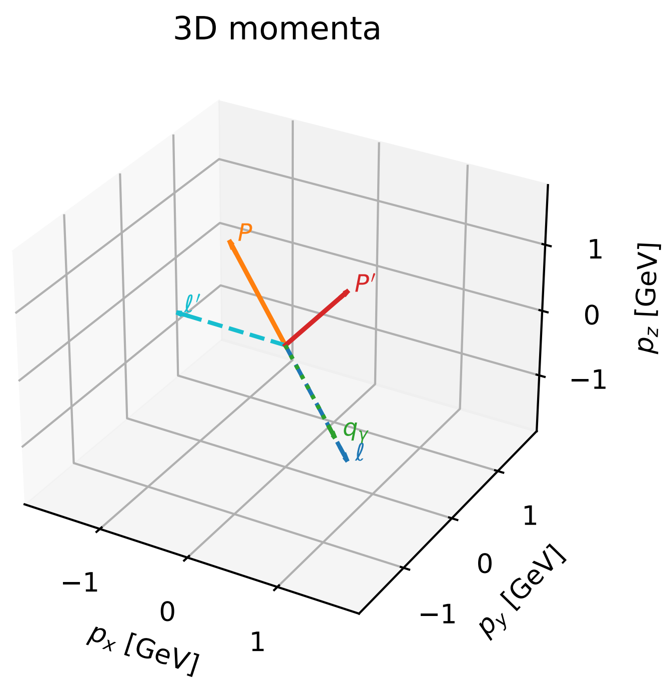
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 0.0000 p_y= 2.2231 p_z= -0.0000
P E= 2.4129 p_x= -0.0000 p_y= -2.2231 p_z= 0.0000
ℓ' E= 1.8959 p_x= -0.0000 p_y= -1.8930 p_z= 0.0000
P' E= 0.9426 p_x= 0.0000 p_y= 0.0930 p_z= 0.0000
q_γ E= 1.8000 p_x= 0.0000 p_y= 1.8000 p_z= 0.0000






L muon + Ty proton characteristic max P_W configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P'
P
ℓ
q_γ
ℓ'
w_purity_high_egamma_lty_region_1 P_W=0.303885
region: high_Egamma_LTy_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{y,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.24788
θ_{in}=1.57079, ϕ_ℓ=3.92699, ϕ_P=0.785398
E_γ=1.8, ϕ_γ=3.92699
Q^2=9.74367, x_B=0.525542, t=-1.8506, W^2=9.6764, y=0.898927
$\theta(\ell', \gamma)$=136.121 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -1.5720 p_y= -1.5720 p_z= -0.0000
P E= 2.4129 p_x= 1.5720 p_y= 1.5720 p_z= 0.0000
ℓ' E= 1.2774 p_x= 1.2728 p_y= 0.0249 p_z= 0.0000
P' E= 1.5611 p_x= 0.0000 p_y= 1.2479 p_z= 0.0000
q_γ E= 1.8000 p_x= -1.2728 p_y= -1.2728 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=2, retained=99.9%)
h_e=-1, h_p=-1, h_γ=+1
muon L, proton Ty
phase=-2.18, Re=-9.29e+00, Im=-1.34e+01
h_e=-1, h_p=+1, h_γ=+1
muon L, proton Ty
phase=-2.87, Re=-5.02e+00, Im=-1.39e+00
ensemble-weighted final-state |A|^2 fraction

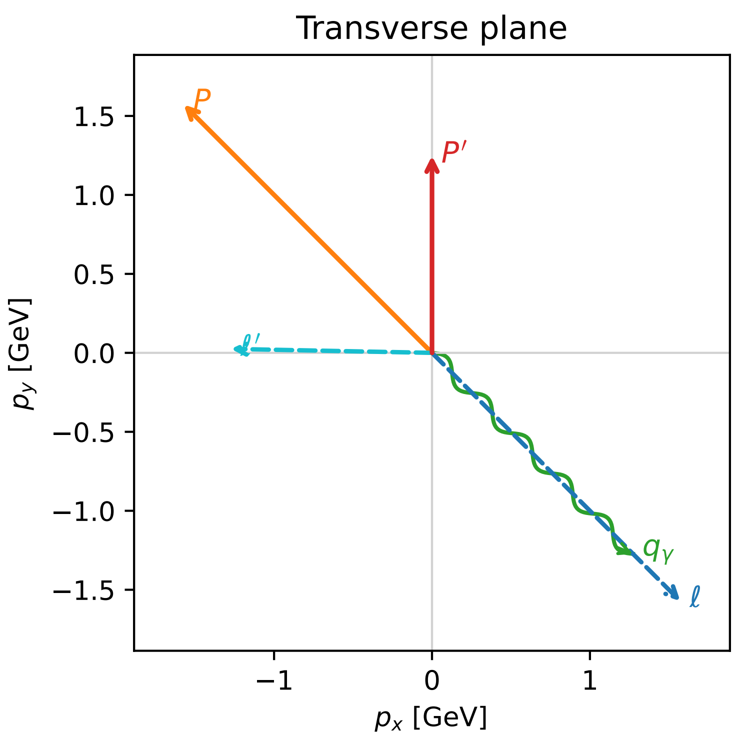
L muon + Ty proton characteristic max P_W configuration

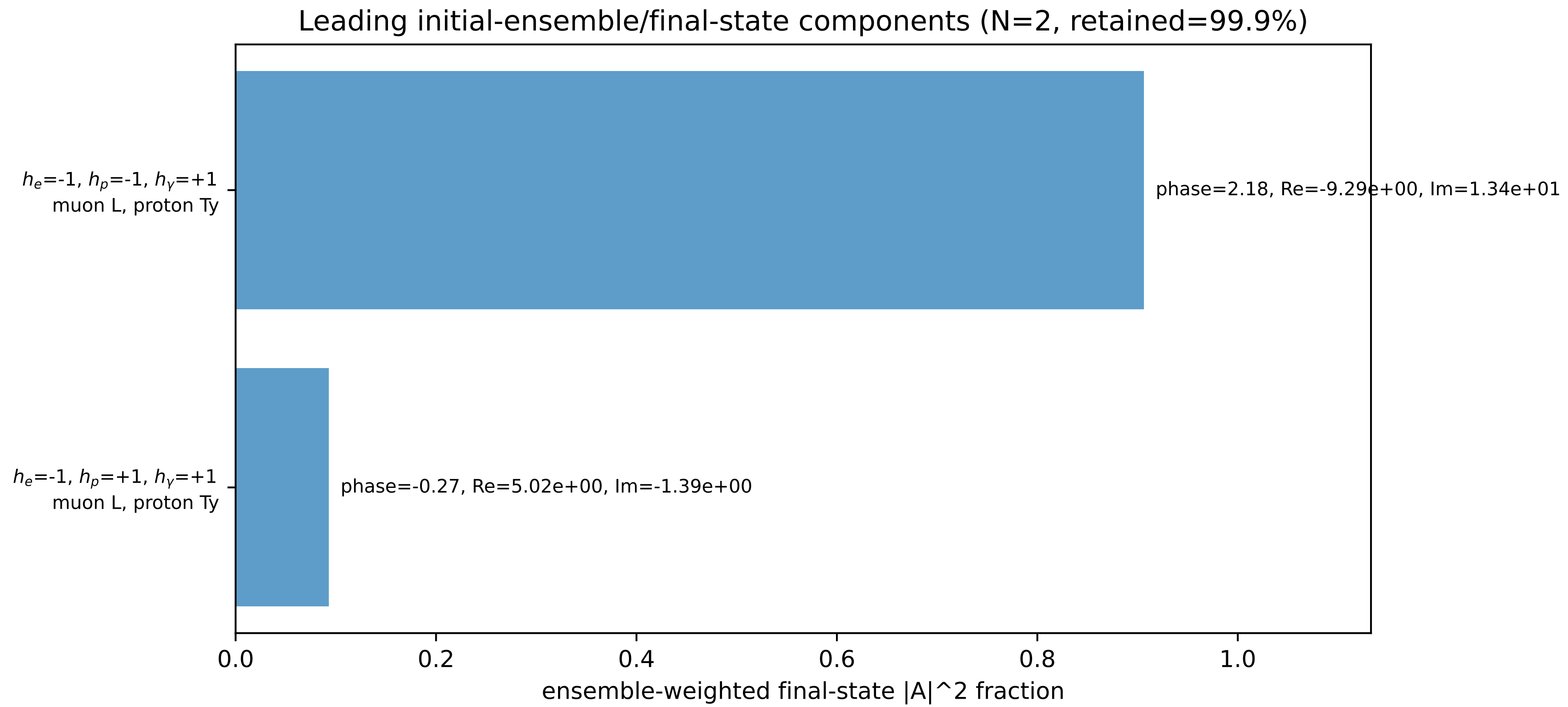


w_purity_high_egamma_lty_region_2 P_W=0.30037
region: high_Egamma_LTy_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{y,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.24788
θ_{in}=1.57079, ϕ_ℓ=5.49779, ϕ_P=2.35619
E_γ=1.8, ϕ_γ=5.49779
Q^2=9.74367, x_B=0.525542, t=-1.8506, W^2=9.6764, y=0.898927
$\theta(\ell', \gamma)$=136.121 deg

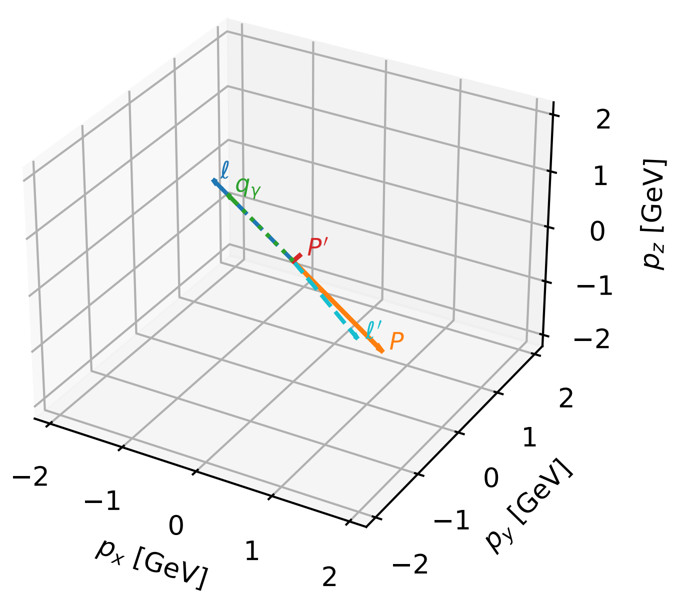
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 1.5720 p_y= -1.5720 p_z= -0.0000
P E= 2.4129 p_x= -1.5720 p_y= 1.5720 p_z= 0.0000
ℓ' E= 1.2774 p_x= -1.2728 p_y= 0.0249 p_z= 0.0000
P' E= 1.5611 p_x= 0.0000 p_y= 1.2479 p_z= 0.0000
q_γ E= 1.8000 p_x= 1.2728 p_y= -1.2728 p_z= 0.0000





L muon + Ty proton characteristic max P_W configuration

3D momenta

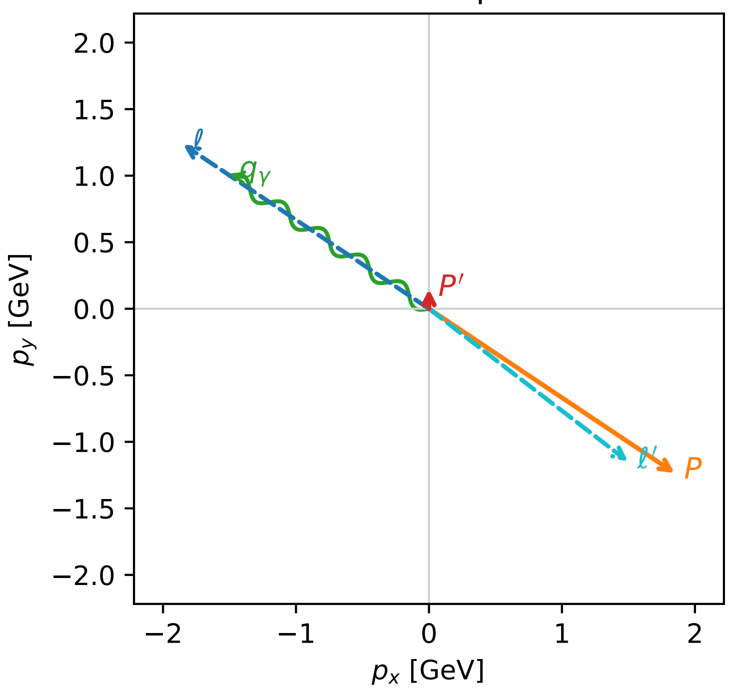


w_purity_high_egamma_lty_region_3 P_W=0.281216
region: high_Egamma_LTy_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{y,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=0.147625
θ_{in}=1.57079, ϕ_ℓ=2.55254, ϕ_P=5.69414
E_γ=1.8, ϕ_γ=2.55254
Q^2=16.7538, x_B=0.842878, t=-3.18728, W^2=4.00295, y=0.963737
$\theta(\ell', \gamma)$=176.268 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -1.8484 p_y= 1.2351 p_z= -0.0000
P E= 2.4129 p_x= 1.8484 p_y= -1.2351 p_z= 0.0000
ℓ' E= 1.8890 p_x= 1.4966 p_y= -1.1477 p_z= 0.0000
P' E= 0.9495 p_x= 0.0000 p_y= 0.1476 p_z= 0.0000
q_γ E= 1.8000 p_x= -1.4966 p_y= 1.0000 p_z= 0.0000

Transverse plane



Leading initial-ensemble/final-state components (N=2, retained=100.0%)